

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1719

Pentland 138 kV
July 01, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1719. The Interconnection Customer proposes a 219.6 MW battery energy storage facility interconnecting at the Pentland 138 kV substation in the Virginia area. The request was received on April 13, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 565 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$31,295,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: White Pine-Cushaw 138 KV Line
Contingency: Pentland Substation-Reusens 138 KV
Loading: 119.0% (Rating: 200 MVA)
DFAX: 100.0%
Pre-project loading: 9.5%
2. Facility: Way Dam-Plaquemine Cogeneration Plant 138 KV line
Contingency: Pentland-Reusens 138 KV L
Loading: 117.8% (Rating: 195 MVA)
DFAX: 100.0%
Pre-project loading: 5.3%
3. Facility: Isla Grande Gis 11191120-Devil'S Hill 138 KV
Contingency: Reusens Substation-Pentland 138 KV Line
Loading: 117.5% (Rating: 211 MVA)
DFAX: 100.0%
Pre-project loading: 13.6%
4. Facility: Isla Grande Gis 11191120-Snowden 138 KV L
Contingency: Reusens-Pentland 138 KV
Loading: 117.4% (Rating: 211 MVA)
DFAX: 100.0%
Pre-project loading: 13.5%
5. Facility: Devil'S Hill-White Pine 138 KV line
Contingency: Reusens-Pentland 138 KV
Loading: 117.2% (Rating: 208 MVA)
DFAX: 100.0%
Pre-project loading: 11.4%

6. Facility: Franklin-Elk Grove-Banning 138 KV
Contingency: Franklin-Elk Grove-S Texa 138 KV
Loading: 114.2% (Rating: 215 MVA)
DFAX: 10.6%
Pre-project loading: 103.4%
7. Facility: Way Dam-Cushaw 138 KV
Contingency: Reusens-Pentland 138 KV
Loading: 114.0% (Rating: 211 MVA)
DFAX: 100.0%
Pre-project loading: 9.9%
8. Facility: Desert Outpost Substation-Banning 138 KV
Contingency: Franklin-Elk Grove-South Texa 138 KV
Loading: 110.4% (Rating: 234 MVA)
DFAX: 10.6%
Pre-project loading: 100.4%
9. Facility: Coal Mtn Substation-Grassmere 138 KV
Contingency: Bear-Clover 345 KV
Loading: 109.4% (Rating: 444 MVA)
DFAX: 5.2%
Pre-project loading: 106.8%
10. Facility: Coal Mtn-Central Austin 138 KV L
Contingency: Clover-Bear 345 KV
Loading: 109.1% (Rating: 453 MVA)
DFAX: 5.2%
Pre-project loading: 106.6%
11. Facility: Central Austin-Waynesboro 138 KV line
Contingency: Bear-Clover 345 KV
Loading: 108.7% (Rating: 465 MVA)
DFAX: 5.2%
Pre-project loading: 106.2%
12. Facility: Alderwood-Waynesboro 138 KV
Contingency: Bear-Clover 345 KV line
Loading: 107.7% (Rating: 496 MVA)
DFAX: 5.2%
Pre-project loading: 105.4%
13. Facility: S Texa Substation-Franklin-Elk Grove 138 KV
Contingency: Desert Outpost Substation-South Texa 138 KV
Loading: 106.4% (Rating: 401 MVA)
DFAX: 16.1%
Pre-project loading: 97.6%
14. Facility: Aibonito Pds-S Crk 138 KV L
Contingency: Bear-Clover 345 KV
Loading: 104.1% (Rating: 667 MVA)
DFAX: 5.7%
Pre-project loading: 102.3%
15. Facility: Intel-S Crk 138 KV Line
Contingency: Aibonito Pds Substation-S Crk 138 KV L
Loading: 103.1% (Rating: 341 MVA)
DFAX: 5.2%
Pre-project loading: 99.7%
16. Facility: Birchwood-Leavitt 345 KV Line
Contingency: Bear Island-Dunn Siding 345 KV

Loading: 101.2% (Rating: 3,031 MVA)
 DFAX: 15.0%
 Pre-project loading: 100.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
White Pine-Cushaw 138 KV Line	Rebuild line with higher-capacity conductor	\$9,390,000	100% (\$9,390,000)
Plaquemine Cogeneration Plant-Wa Dam 138 KV L	Rebuild line with higher-capacity conductor	\$1,992,000	100% (\$1,992,000)
Isla Grande Gis 11191120-Devil'S Hill 138 KV	Rebuild line with higher-capacity conductor	\$4,860,000	100% (\$4,860,000)
Isla Grande Gis 11191120-Snowden 138 KV L	Rebuild line with higher-capacity conductor	\$3,474,000	100% (\$3,474,000)
Devil'S Hill-White Pine 138 KV	Rebuild line with higher-capacity conductor	\$4,931,000	100% (\$4,931,000)
Franklin-Elk Grove-Banning 138 KV	Rebuild line with higher-capacity conductor	\$2,684,000	9% (\$248,000)
Way Dam-Cushaw 138 KV	Rebuild line with higher-capacity conductor	\$2,676,000	100% (\$2,676,000)
Banning-Desert Outpost 138 KV L	Rebuild line with higher-capacity conductor	\$13,976,000	9% (\$1,290,000)
Coal Mtn-Grassmere 138 KV	Reconductor line	\$1,387,000	15% (\$204,000)
Coal Mtn-Central Austin 138 KV L	Reconductor line	\$2,044,000	15% (\$301,000)
Central Austin-Waynesboro 138 KV	Reconductor line	\$917,000	15% (\$135,000)
Alderwood-Waynesboro 138 KV	Reconductor line	\$2,015,000	15% (\$296,000)
Franklin-Elk Grove-S Texa 138 KV	Reconductor line	\$3,544,000	12% (\$442,000)
Aibonito Pds-S Crk 138 KV L	Reconductor line	\$2,080,000	4% (\$79,000)
Intel-S Crk 138 KV Line	Reconductor line	\$3,234,000	6% (\$187,000)
Leavtt-Birchwood 345 KV	Reconductor line	\$64,304,000	1% (\$790,000)
	Total allocated to project		\$31,295,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.